When Demographic Labels Substitute for Mechanisms: Misclassification of Exposure Pathways and Susceptibility in Environmental Health Research

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In research on climate-related environmental exposures and health outcomes, the identification of susceptible populations frequently relies on demographic stratification (i.e., sex, age, race, and socioeconomic status) without adequate consideration of the mechanisms generating observed group differences. Analytical frameworks typically emphasize climate exposure, demographic and socioeconomic variables, while overlooking other factors that shape differential exposure. This approach risks obscuring the exposure pathways, thereby conflating socially patterned exposure with intrinsic biological sensitivity. The association between air pollution and coronary heart disease provides a salient example.

A large amount of epidemiological evidence indicates a robust association between air pollution and cardiovascular disease (Lee et al., 2014). In studies examining the effects of wildfire smoke or PM2.5 on coronary heart disease, most findings report higher relative risks among women than men. However, these analyses frequently do not fully consider the sex-specific differences in exposure.

For example, Baumgartner et al. (2014) reported in rural China that black carbon exposure was significantly associated with elevated blood pressure in women, with stronger associations among those residing near highways. This pattern likely reflects gendered divisions of labor: women may be more frequently engaged in charcoal or waste burning activities, particularly in areas with weaker environmental enforcement, resulting in higher cumulative exposure. In contrast, Weichenthal et al. (2014) observed in rural Iowa and North Carolina that PM2.5 was associated with cardiovascular mortality in men, but no similar association was observed in women. The association among men may reflect occupational exposure, as men in these settings are more likely to participate in prolonged outdoor agricultural work and therefore experience higher exposure to diesel exhaust and ambient particulates.

Taken together, these findings suggest that observed sex differences in cardiovascular risk may arise from socially structured patterns of exposure rather than inherent biological sensitivity. This raises a critical hypothesis: higher reported risks of coronary heart disease among women in some air pollution studies may partially reflect greater exposure to indoor air pollution from cooking and other domestic tasks, rather than physiological vulnerability. However, most current studies do not explicitly integrate detailed exposure pathways into their analytical frameworks, resulting in observed sex differences frequently being interpreted as evidence of differential biological sensitivity, rather than as outcomes of socially structured exposure disparities. This analytical limitation risks reifying demographic categories as biological determinants and obscuring the social and behavioral mechanisms that generate health inequities.

References

Baumgartner, J., Zhang, Y., Schauer, J., Wang, Y., Huang, W., & Ezzati, M. (2014). Highway proximity and Black Carbon from cookstoves as a risk factor for higher blood pressure in rural China. *ISEE Conference Abstracts*, 2014(1). https://doi.org/10.1289/isee.2014.o-185

Lee, B.-J., Kim, B., & Lee, K. (2014). Air Pollution Exposure and cardiovascular disease. *Toxicological Research*, 30(2), 71–75. https://doi.org/10.5487/tr.2014.30.2.071

Weichenthal, S., Villeneuve, P. J., Burnett, R. T., van Donkelaar, A., Martin, R. V., Jones, R. R., DellaValle, C. T., Sandler, D. P., Ward, M. H., & Hoppin, J. A. (2014). Long-term exposure to fine particulate matter: Association with Nonaccidental and cardiovascular mortality in the Agricultural Health Study Cohort. *Environmental Health Perspectives*, 122(6), 609–615. https://doi.org/10.1289/ehp.1307277